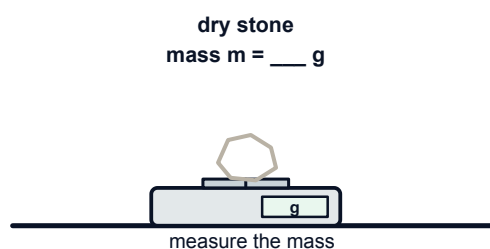


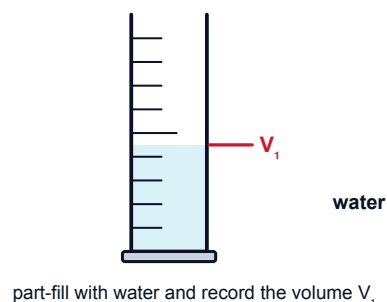
Density of an irregular solid

A student measures the density of a small stone.

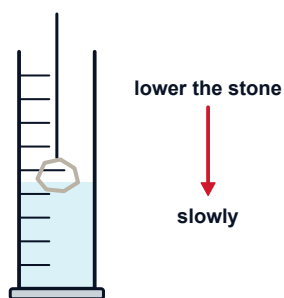
- 1** Measure the mass of the dry stone with a balance.



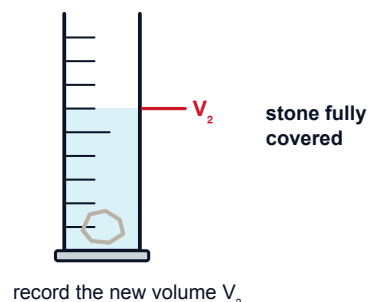
- 2** Part-fill a measuring cylinder with water and record the volume.



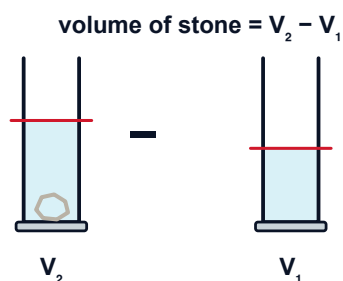
- 3** Lower the stone into the water until it is fully covered.



- 4** Record the new volume shown on the measuring cylinder.



- 5** Subtract the two volumes to find the volume of the stone.



- 6** Calculate the density: mass $\div$ volume.

$$\rho = m \div V$$

density (g/cm^3) = mass (g) $\div$ volume (cm^3)

$$m = \text{mass of the stone, } V = V_2 - V_1$$